

Vyomini Joshi

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PROFILE

Computer Science undergraduate at NIT Surat with a strong foundation in Data Structures Algorithms, full-stack web development, and AI-powered applications. Passionate about building scalable software solutions and solving algorithmic problems. Solved **300+ coding problems** on **LeetCode** with a peak rating of **1526**, demonstrating strong problem-solving and analytical skills.

EDUCATION

National Institute of Technology, Surat

2024 – 2028

B.Tech – Computer Science and Engineering (CGPA: 8.42 Till 4th Semester)

Surat, India

- **Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems, Information Security
- **Academic Highlights:** CBSE Class XII: 94.4% | CBSE Class X: 95.6%

SKILLS

Languages : C++, C, Python, JavaScript, SQL (MySQL, PostgreSQL)

Systems & Networking: Computer Networks, TCP/IP, Socket Fundamentals, Linux Shell

Backend & Web: FastAPI, Node.js, Express.js, React.js, Tailwind CSS, REST APIs

Databases & Tools: PostgreSQL, MySQL, MongoDB, Firebase, Git, GitHub, VS Code

Libraries & Signal Processing: NumPy, Pandas, Scikit-learn, MediaPipe, OpenCV Basics

PROJECTS

CaseCraftAI | AI-Powered Legal Documentation Platform

- **Tech:** Python, docxtempl, PostgreSQL
- Developed an AI-powered document generation module supporting **4+ legal document types**, including Seizure Receipts, Remand Requests, Medical Treatment Letters, and Accused Panchanamas.
- Implemented dynamic docxtempl templates by mapping **30+ case fields** into document templates for automated generation.
- Integrated the module with the backend, enabling one-click generation of downloadable legal documents directly from case records.

CardioVision | Browser-Based Contactless Heart Rate Monitor

- **Tech:** JavaScript, MediaPipe, Signal Processing, HTML5, CSS3
- Developed a browser-based heart rate monitoring system processing webcam video at approximately **30 FPS** using remote photoplethysmography (rPPG).
- Applied MediaPipe Face Mesh (**468 facial landmarks**) to extract facial ROIs and implemented the CHROM algorithm for pulse signal estimation.
- Performed frequency-domain analysis using DFT to compute live BPM values and enabled JSON export for recorded sessions.

Rabuste Café | Coffee Ordering Platform

- **Tech:** React.js, Tailwind CSS, JavaScript, HTML/CSS
- Developed a responsive frontend comprising **8+ reusable React components** for menu browsing, shopping cart, and checkout workflows.
- Designed a relational database schema with **4+ core entities** including users, menu items, carts, and orders.
- Integrated an LLM API to deliver AI-powered menu recommendations and customer assistance within the ordering interface.

ACHIEVEMENTS

- **Qualified** the Round 1 CTF of **E-Rakshak**, a cybersecurity hackathon by Surat City Police and SVNIT, and advanced to the mentored project round (Aug. 2026).
- Won **Google Winter of Code 2026**
- Finalist at **Web Wonders 2025**
- Secured **99.44 percentile** in JEE Main 2024
- Awarded the **INSPIRE-MANAK Scholarship** by the Government of India
- Selected as a **National-Level Participant** for the Jawaharlal Nehru National Science, Mathematics and Environment Exhibition

POSITIONS OF RESPONSIBILITY

- **Student Representative** – Nexus, NIT Surat *Aug 2025 – Present*
Designed presentations for technical events and mentored **10+ junior students** in web development fundamentals.
- **Executive Member** – E-Cell, NIT Surat *Sep 2025 – Feb 2026*
Prepared official documentation, permission letters, and financial correspondence supporting the organization of entrepreneurship events.
- **NSS Volunteer** *May 2025 – Jan 2026*
Contributed to community service through cleanliness drives, social awareness campaigns, and physical fitness initiatives.